

Advanced Python Challenge

Question: Student Performance & Statistical Analysis

You are given the following list containing the marks of 15 students:

```
marks = [72, 85, 91, 68, 85, 77, 93, 56, 72, 88, 95, 64, 79, 91, 83]
```

Write a Python program that performs the following operations.

Part 1 — List Processing

1. Display the original list.
2. Sort the marks in **ascending order**.
3. Sort the marks in **descending order**.
4. Remove duplicate marks using a **set**.
5. Convert the unique marks back into a list and sort them in ascending order.

Part 2 — Mathematical Calculations

Using the original marks, calculate:

6. Total marks
7. Average
8. Highest mark
9. Lowest mark
10. Range

Part 3 — Advanced Statistics

Calculate the **variance** using:

$$Variance = \frac{\sum (x - \bar{x})^2}{n}$$

where:

- x = each student's mark
- $\bar{x}$ = average
- n = number of students

Then calculate the **standard deviation**:

$$SD = \sqrt{Variance}$$

Important: Use Python's `math.sqrt()` for the square root.